

Assignment 4: Text and Sequence Data

Introduction

This assignment uses the IMDB dataset to implement the techniques of deep learning on text data. The goal is to compare pre trained GloVe embeddings and trainable embeddings and compare their performance at various training sample sizes as it has been discussed in Chapter 11.

Methodology

The IMDB data were subjected to the following parameters: the maximum number of words in a review was set to 150, vocabulary was set to the top 10,000 words, and a validation sample of 10,000 was utilized. Two models were created: GloVe pretrained and trainable embeddings. The performance was assessed on different training sizes (100 to 5000 samples).

Results and Discussion

The best model witnessed in this experiment was the trainable embedding model, which was trained on 5000 samples with a validation accuracy of 0.7790.

On smaller datasets (1001000 samples) trainable embeddings were always worse than pretrained ones. This implies that pretrained embeddings can be used with limited data because they already have knowledge of large text corpora.

Nevertheless, the trainable embedding model was getting better as the sample size in the training increased. The trainable embedding model started to be more successful than the pretrained model at around 2000 samples. This benefit persisted with increased data sets.

The findings of this study indicate one of the most important tradeoffs since the pretrained embeddings prove useful when data is limited, and trainable embeddings are more efficient when there is enough data. These results are consistent with Chapter 11 ideas of representation learning.

Conclusion

The results of this experiment prove that the model performance is a factor of the availability of data. Pretrained embeddings work well when the amount of data is small, whereas trainable embeddings work well in large datasets. A significant finding to consider the model selection was that the crossover point was found at around 2000 samples.

Graph

